

KONGJU NATIONAL UNIVERSITY

Particle Filters

Nonparametric Localization and Resampling Experiments

Week 5 Assignment

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PROGRAM

M.S. in Mechanical Engineering

COURSE

Autonomous Driving

GitHub: https://github.com/Chal29/Autonomous/tree/main/Week5%20assignment

Q1. Effect of Particle Count on Localization Performance

Objective and Method

I investigated how particle count affects global localization reliability. In a 10 × 10 m environment with four landmarks, I tested M = 100, 500, 2,000 and 10,000 particles, each over 20 random seeds and 25 steps. Range noise was 0.35 m; low-variance resampling occurred when ESS < M/2. Success meant final Euclidean error ≤ 0.5 m.

M	Success	Rate	Mean error
100	6/20	30%	4.597 m
500	14/20	70%	1.338 m
2,000	20/20	100%	0.175 m
10,000	18/20	90%	0.212 m

Discussion

Success improved from 30% with 100 particles to 100% with 2,000, consistent with improved state-space coverage and reduced convergence to incorrect hypotheses. The 10,000-particle case achieved 90%, but the decrease over only 20 stochastic trials does not demonstrate that additional particles harm performance. More repetitions would be needed to assess this difference.

Conclusion

Among the tested configurations, 2,000 particles gave the best observed trade-off: 100% success and 0.175 m mean final error. Additional particles did not improve these measured outcomes.

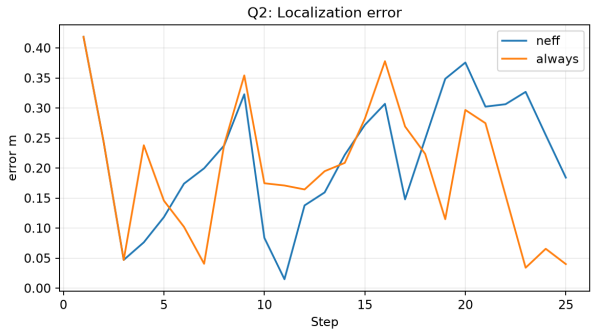
Q2. Comparison of Resampling Strategies

Objective and Method

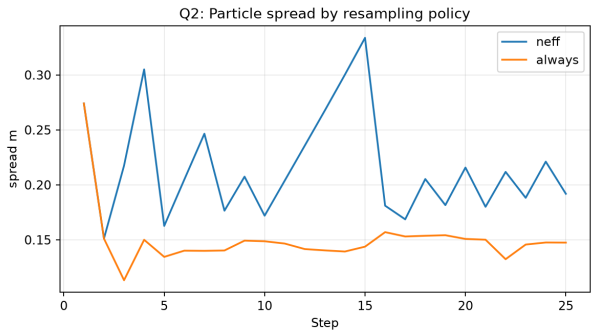
Using M = 2,000 particles for 25 steps, I compared resampling every step with adaptive resampling when ESS < M/2 = 1,000.

Results

Adaptive ESS resampling occurred 11 times versus 25 times for always resampling, a 56% reduction. Mean errors were 0.221 m (ESS) and 0.195 m (always); final errors were 0.184 m and 0.040 m. Mean particle spreads were 0.216 m and 0.150 m, respectively.



Localization error over 25 steps.



Particle spread under both policies.

Discussion and Conclusion

Both strategies localized successfully. Always resampling achieved a lower final error and a tighter particle cloud, but frequent selection may eliminate low-weight hypotheses. ESS-based resampling retained a broader spread while reducing operations. Spread alone does not prove useful diversity, and a single trial cannot establish overall superiority. Adaptive resampling offered acceptable accuracy with fewer resampling operations.

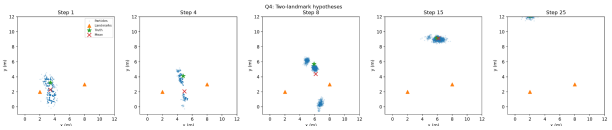
Q3. Precise-Sensor Paradox and Particle Depletion

Objective and Method

With M = 2,000, I reduced range noise from σ = 0.35 m to 0.05 m, then tested likelihood tempering and increased motion noise as remedies.

Results

Baseline final error was 0.184 m; the unmodified precise-sensor filter failed at 7.527 m. Its average ESS was 1.07 and its final particles descended from one ancestor. Tempering reduced final error to 0.051 m; increased motion noise reduced it to 0.031 m.



Precise-sensor failure and two recovery approaches.

Discussion and Conclusion

A narrow likelihood gave almost all predicted particles negligible weight, causing weight collapse and resampling-driven depletion around an incorrect pose. The particle snapshots corroborated divergence. Tempering reduced excessive likelihood concentration; additional motion noise improved exploration. Sensor precision alone cannot guarantee accuracy when the proposal distribution lacks particles near the true state.

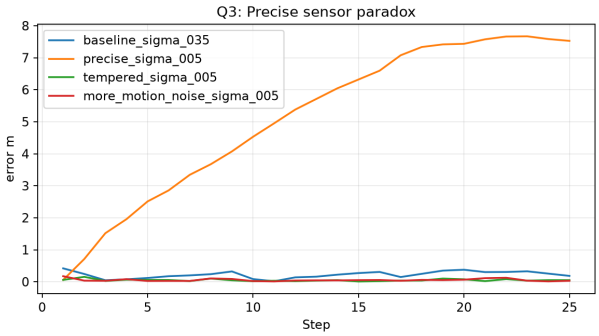
Q4. Two-Landmark Ambiguity and Multimodal Localization

Objective and Method

I restricted the filter to two range landmarks and inspected particle distributions at steps 1, 4, 8, 15 and 25.

Results

Multiple pose hypotheses appeared initially. Localization error was 2.039 m at step 4 and 1.364 m at step 8, then decreased to 0.239 m at step 15 and 0.245 m at step 25.



Two-landmark hypotheses at five time steps.

Discussion and Conclusion

Two distance circles can intersect at two geometrically valid positions, producing separated particle clusters. Their weighted mean may lie between clusters and poorly represent either hypothesis. As motion and sequential measurements accumulated, inconsistent hypotheses lost support and particles concentrated near the true path. The final 0.245 m error indicates successful localization in this run.

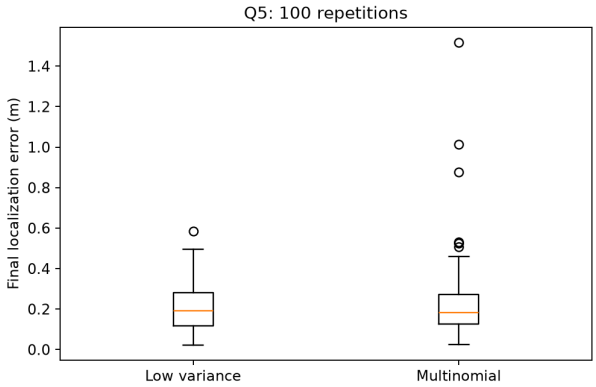
Q5. Comparison of Low-Variance and Multinomial Resampling

Objective and Method

I compared systematic (low-variance) and multinomial resampling across 100 simulations using final localization-error statistics and a boxplot.

Results

Mean final errors were 0.215 m (systematic) and 0.230 m (multinomial). Sample error variances were 0.0131 m² and 0.0418 m²: 68.7% lower for systematic resampling. Median errors were 0.192 m and 0.182 m; maximum errors were 0.584 m and 1.518 m, respectively.



Final-error distribution over 100 repetitions.

Discussion and Conclusion

Multinomial sampling independently draws ancestors and may overduplicate some particles. Systematic sampling uses evenly spaced positions with a single random offset, limiting selection randomness. Median accuracy was similar, but multinomial resampling had more severe outliers. In this experiment, systematic resampling produced more consistent localization; this does not imply it always minimizes error.

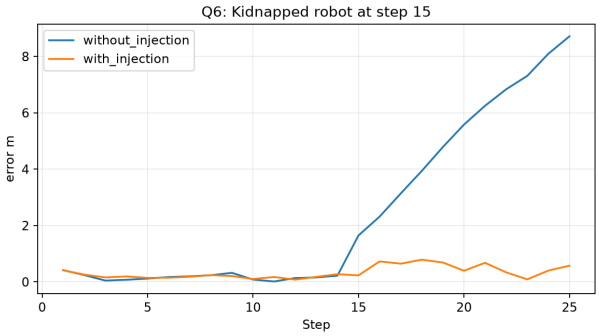
Q6. Kidnapped Robot Problem and Random Particle Injection

Objective and Method

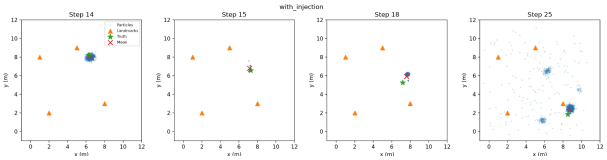
I teleported the robot at step 15 and compared a filter without injection to one that introduced random particles, using position-error curves and particle snapshots.

Results

Without injection, error reached 3.939 m at step 18 and 8.714 m at step 25. With injection, errors were 0.789 m at step 18, 0.090 m at step 23 and 0.576 m at step 25: a 93.4% reduction in final error.



Localization error after relocation at step 15.



Particle snapshots with random injection.

Discussion and Conclusion

Without injection, particles stayed near an obsolete pose hypothesis and failed to recover. Injected particles explored alternative locations and enabled relocalization near the true position. The post-recovery error still fluctuated, so tracking was not perfectly stable. Random particle injection substantially improved recovery from unexpected relocation in this trial.

Q6. Particle distributions and recovery (continued)

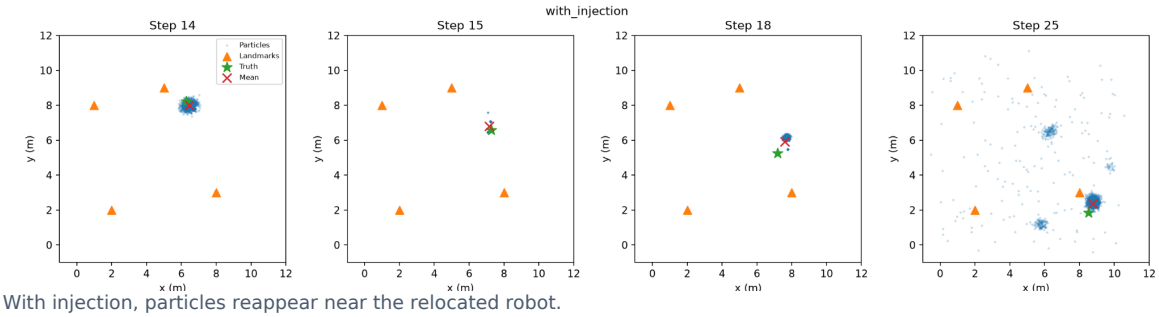


Figure 6. With injection, particles reappear near the relocated robot.

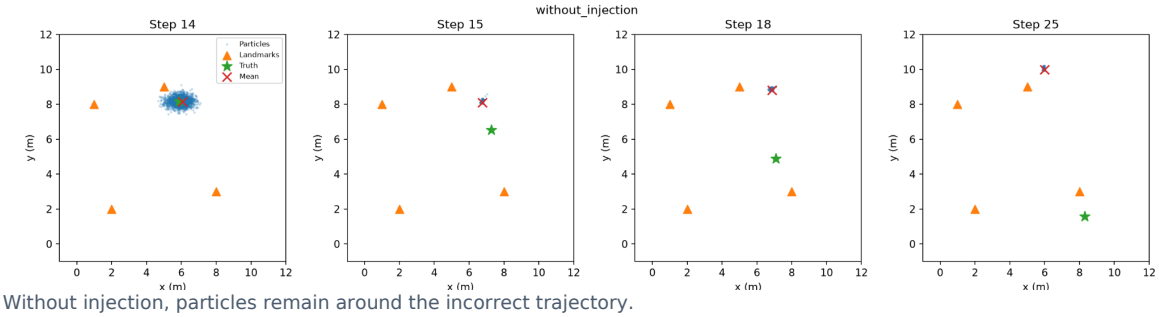


Figure 7. Without injection, particles remain around the incorrect trajectory.

The injected filter reached 0.090 m at step 23 and finished at 0.576 m; without injection, error reached 8.714 m. Injection enabled relocalization, although error still fluctuated.

Q7. Binary Bayes filter: log-odds and overconfidence

Step 1 | Define the posterior belief and its log odds.

$$b_t = P(x \mid z_{1:t}), \qquad \ell_t = \ln\left(\frac{b_t}{1-b_t}\right)$$

Step 2 | Apply Bayes rule to obtain the posterior-odds update.

$$\frac{b_t}{1-b_t} = \frac{b_{t-1}}{1-b_{t-1}} \frac{P(z_t \mid x)}{P(z_t \mid \neg x)}$$

Step 3 | Take logarithms to convert multiplication into addition.

$$\ell_t = \ell_{t-1} + \ln\left(\frac{P(z_t \mid x)}{P(z_t \mid \neg x)}\right)$$

Step 4 | Substitute the inverse sensor model (lecture Eq. 5.5).

$$\ell_t = \ell_{t-1} + \ln\left(\frac{P(x \mid z_t)}{1-P(x \mid z_t)}\right) - \ln\left(\frac{P(x)}{1-P(x)}\right)$$

Recover the probability with the logistic transformation:

$$b_t = \frac{1}{1+e^{-\ell_t}}$$

Why correlated measurements cause overconfidence

If the same observation is mistakenly treated as k independent measurements, its log-odds increment is added k times. This counts the same evidence repeatedly.

$$\ell_k = \ell_0 + k\,\Delta\ell, \qquad \Delta\ell = \ln(0.8/0.2) = \ln 4$$

Example: starting from P(x) = 0.5, repeating evidence with inverse-sensor probability 0.8 five times yields P(x) = 4^5/(1 + 4^5) = 0.9990. The 99.9% confidence is unjustified because the observations were not independent.